

Quiz — 1.4.3 Boolean Algebra

Name: _____ Date: _ **Mark:** ____ / 30

OCR H446 · Component 01 · 1.4.3 — show all working

Section A — Multiple choice (8 marks)

1. For which input combination does a 2-input **AND** gate output 1? A. A=0, B=0 B. A=1, B=0 C. A=0, B=1 D. A=1, B=1 Your answer: ☐
 2. Which logic gate outputs 1 **only when its two inputs differ**? A. AND B. OR C. XOR D. NOR Your answer: ☐
 3. Which two gates are described as **universal** (any circuit can be built from them alone)? A. AND and OR B. NAND and NOR C. XOR and NOT D. OR and NOT Your answer: ☐
 4. De Morgan's law states that $\neg(A + B)$ is equivalent to: A. $\neg A + \neg B$ B. $\neg A \cdot \neg B$ C. $A \cdot B$ D. $\neg(A \cdot B)$ Your answer: ☐
 5. Which Boolean identity is $\neg(\neg A) = A$? A. Idempotent B. Absorption C. Double negation D. Complement Your answer: ☐
 6. In a **half adder**, the Sum output is given by: A. $A \cdot B$ B. $A + B$ C. $A \oplus B$ D. $\neg(A \cdot B)$ Your answer: ☐
 7. What is the role of a **D-type flip-flop**? A. To add two bits together B. To store one bit of data on a clock edge C. To invert its input D. To convert binary to hexadecimal Your answer: ☐
 8. When grouping 1s on a Karnaugh map, groups must be: A. Always exactly size 3 B. As small as possible C. Powers of two (1, 2, 4, 8...) and as large as possible D. Only ever a single cell Your answer: ☐
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Section B — Calculations & short answer (17 marks)

9. Complete the truth table for the expression $Q = \neg(A \cdot B) + (A \cdot C)$. [5]

A	B	C	$A \cdot B$	$\neg(A \cdot B)$	$A \cdot C$	Q
0	0	0				
0	0	1				
0	1	0				
0	1	1				
1	0	0				
1	0	1				
1	1	0				
1	1	1				

10. Use **De Morgan's law** (and double negation) to simplify $\neg(\neg A + B)$ to an expression with no outer bar. Show each step. **[3]**

Working space:

11. A **half adder** receives inputs **A = 1** and **B = 1**. State the **Sum** and **Carry** outputs and the Boolean expression for each. **[3]**

Working space:

12. Complete this truth table for a **half adder** for all input combinations. **[3]**

A	B	Sum ($A \oplus B$)	Carry ($A \cdot B$)
0	0		
0	1		
1	0		
1	1		

13. Simplify the expression $A \cdot (A + B)$ using a Boolean identity, and name the law used. **[3]**

Working space:

Section C — Exam-style question (5 marks)

14. A **full adder** adds three input bits: A, B and a carry-in (C_{in}), producing a Sum and a carry-out (C_{out}).

(a) Give the Boolean expression for the **Sum** output of a full adder. **[1]**

(b) Give the Boolean expression for the **Cout** output of a full adder. **[2]**

(c) Explain how a full adder differs from a half adder, and why full adders are needed to add multi-bit binary numbers. **[2]**

Working space:

Answer key

Section A (8 marks — 1 each) 1. **D** — AND outputs 1 only when both inputs are 1. 2. **C** — XOR outputs 1 when inputs differ. 3. **B** — NAND and NOR are the universal gates. 4. **B** — $\neg(A + B) = \neg A \cdot \neg B$ (break the bar, swap OR $\rightarrow$ AND). 5. **C** — double negation. 6. **C** — half adder Sum = $A \oplus B$. 7. **B** — D-type flip-flop stores one bit on a clock edge. 8. **C** — K-map groups are powers of two and as large as possible.

Section B

9. [5] $Q = \neg(A \cdot B) + (A \cdot C)$:

A	B	C	$A \cdot B$	$\neg(A \cdot B)$	$A \cdot C$	Q
0	0	0	0	1	0	1
0	0	1	0	1	0	1
0	1	0	0	1	0	1
0	1	1	0	1	0	1
1	0	0	0	1	0	1
1	0	1	0	1	1	1
1	1	0	1	0	0	0
1	1	1	1	0	1	1

Mark: $A \cdot B$ column correct (1); $\neg(A \cdot B)$ column correct (1); $A \cdot C$ column correct (1); Q correct for rows 1–4 (1); Q correct for rows 5–8 — note only row $A=1, B=1, C=0$ gives $Q=0$ (1).

10. [3] Simplify $\neg(\neg A + B)$: - Apply De Morgan's to the outer bar: $\neg(\neg A + B) = \neg(\neg A) \cdot \neg B$ (break the bar, change + to $\cdot$, negate each term). (1) - Apply double negation: $\neg(\neg A) = A$. (1) - Result: $A \cdot \neg B$. (1)

11. [3] Half adder with $A = 1, B = 1$: - Sum = $A \oplus B = 1 \oplus 1 = 0$ (1) - Carry = $A \cdot B = 1 \cdot 1 = 1$ (1) - Correct expressions stated (Sum = $A \oplus B$, Carry = $A \cdot B$) (1). (So $1 + 1 =$ binary 10: Sum 0, Carry 1.)

12. [3] Half adder truth table:

A	B	Sum ($A \oplus B$)	Carry ($A \cdot B$)
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Mark: Sum column correct (1); Carry column correct (1); fully correct table with $1+1$ giving Sum 0 Carry 1 (1).

13. [3] $A \cdot (A + B)$: - This is the **Absorption law**: $A \cdot (A + B) = A$. (1 for naming Absorption) - Result: A . (1) - Justification (1): expand $A \cdot (A+B) = A \cdot A + A \cdot B = A + A \cdot B$ (idempotent $A \cdot A = A$); then $A + A \cdot B = A$ (A absorbs $A \cdot B$), so the whole expression reduces to A .

Section C

14. [5] (a) [1] $\text{Sum} = A \oplus B \oplus \text{Cin}$.

(b) [2] $\text{Cout} = (A \cdot B) + (\text{Cin} \cdot (A \oplus B))$. (1 mark for the $A \cdot B$ term, 1 mark for the $\text{Cin} \cdot (A \oplus B)$ term ORed together. Equivalent correct forms such as $A \cdot B + B \cdot \text{Cin} + A \cdot \text{Cin}$ also gain full marks.)

(c) [2] A **half adder** adds only **two** bits (A and B) and has **no carry-in**, so it cannot accept a carry from a previous column (1). A **full adder** adds **three** bits (A, B and Cin), so full adders can be **chained** (a ripple-carry adder) with each carry-out feeding the next column's carry-in, allowing multi-bit binary numbers to be added correctly (1).

Total: $8 + 17 + 5 = 30$ marks.